

i3 Technologies

Becoming a Web Hosting Provider on IBM Cloud

Detailed Technical Architecture, Operations & Domain Registrar Guide

Built on existing IBM ROKS 4.17 (eu-de) · i3 AI Platform

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This guide shows how to extend the existing i3 AI Platform (namespaces, ArgoCD GitOps, Keycloak, Model Gateway, Postgres HA) into a multi-tenant web-hosting platform for partners and customers, plus the realistic path to domain registration services.

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2. Executive Summary & Strategic Positioning

i3 Technologies already operates a production-grade AI platform on IBM Red Hat OpenShift on IBM Cloud (ROKS 4.17) in the Frankfurt (eu-de) region. The platform uses GitOps (ArgoCD), multi-namespace isolation, Keycloak SSO, Postgres HA (Crunchy PGO), LiteLLM + Ollama model gateway, and a mature secret / monitoring stack.

This document describes how to productise the same infrastructure into a commercial web-hosting offering for partners and end customers, while optionally adding domain registration services. Two realistic paths exist for domains:

- **Domain Reseller (recommended first step)** — Partner with an existing ICANN-accredited registrar (reseller API). Fast to market, lower capital and compliance burden.
- **Full ICANN-accredited Registrar** — Longer (3–9+ months), higher cost (application + capital + escrow + registry deposits), but full control and higher margin.

The technical foundation for hosting itself is already largely present: OpenShift Routes / Ingress, IBM Cloud Internet Services (CIS) for DNS/CDN/WAF/DDoS, Object Storage for static sites, and the ability to spin up isolated namespaces or even dedicated clusters/projects per customer.

3. Current State — i3 AI Platform on IBM ROKS

Reference: existing System Administrator Guide (commit cafb76f). Key assets that will be reused or extended:

Asset	Current Use	Hosting Relevance
ROKS 4.17 (eu-de)	Shared multi-product cluster	Base platform; can host tenant workloads or spawn additional clusters
ArgoCD + GitOps (platform/)	Auto-sync on push to main	Tenant provisioning via Git or ApplicationSets
Namespaces (i3-evalos, i3-pmaas, ...)	Product isolation	Pattern for customer namespaces or dedicated projects
Keycloak (i3-auth) realm i3	SSO for all products	Customer / partner SSO or multi-realm extension
i3-data Postgres HA + PgBouncer	Shared databases	Optional managed DB add-on for customers
Model Gateway + LiteLLM	Shared AI inference	Optional AI features for hosted sites
OpenShift Routes + public URLs	*.i3technologies.co.ke	Customer custom domains via CIS / CNAME
IBM Cloud Object Storage	(if used)	Ideal for static website hosting

Important constraint from current guide: Never touch namespaces solution-01 ... solution-08 (live customer workloads on a separate server). Keep the AI platform and any new multi-tenant hosting plane carefully separated (namespaces, network policies, resource quotas, and preferably separate node pools or clusters).

4. Target Architecture for Multi-Tenant Web Hosting

4.1 High-Level Logical Architecture

Recommended layered model on top of the existing ROKS cluster (or a dedicated sibling ROKS cluster for stronger isolation):

1. **Edge / Delivery layer** — IBM Cloud Internet Services (CIS): authoritative DNS, Global Load Balancer, CDN/caching, WAF, DDoS protection, TLS termination options.
2. **Ingress layer** — OpenShift Ingress Controllers / Routes (or Kubernetes Ingress). Support for custom domains via CNAME or full CIS zone.
3. **Compute layer** — Tenant workloads (static sites, WordPress/PHP, Node, Python, containers) running in isolated namespaces or projects with ResourceQuotas, LimitRanges, NetworkPolicies.
4. **Data layer** — Optional managed Postgres (existing PGO or IBM Cloud Databases), Object Storage for static assets / backups, Redis if needed.
5. **Control plane** — GitOps (ArgoCD ApplicationSets or tenant CRDs), provisioning API / portal, billing metering, Keycloak (or multi-tenant IdP).
6. **Observability** — Existing Prometheus/Grafana + Langfuse pattern extended with per-tenant dashboards and alerts.

4.2 Isolation Models (choose based on customer segment)

Model	Isolation	Cost / Complexity	Best For
Shared namespace + labels	Low (soft)	Lowest	Internal / low-risk demos
Dedicated namespace per tenant	Medium (NetworkPolicy + Quota)	Low–Medium	Most partners & SMB customers
Dedicated project / node pool	High	Medium	Larger customers, regulated
Dedicated ROKS cluster	Highest	Highest	Enterprise / compliance-heavy

Recommendation: Start with **dedicated namespace per tenant** on the existing or a new ROKS cluster, enforce NetworkPolicies (including the DNS egress pattern already documented in the SysAdmin guide), ResourceQuotas, and PodSecurityStandards. Move high-value customers to dedicated node pools or clusters later.

5. Hosting Product Tiers & Service Catalogue

Define clear commercial packages so engineering and sales stay aligned:

Tier	What is included	Typical technology
Static / JAMstack	HTML/CSS/JS, SPA, docs sites, landing pages. Custom domain + TLS + CDN.	IBM COS static website hosting + CIS, or Nginx/Apache on OpenShift + CIS
Managed WordPress / CMS	WordPress, Directus, or headless CMS. Automatic updates, backups, staging.	Containerised WP + MariaDB/Postgres, or existing Directus pattern in i3-ott
Container / App Hosting	Bring-your-own container or Git-based deploy (S2I / BuildConfig). Auto-scale.	OpenShift Deployment + Route + optional HPA; GitOps via ArgoCD
Database Add-on	Managed Postgres (shared or dedicated schema / instance).	Existing PGO cluster or IBM Cloud Databases for PostgreSQL
AI-Enhanced Hosting	Optional LiteLLM / Ollama endpoints, chat widgets, RAG for customer sites.	Existing Model Gateway (with strict rate limits & keys per tenant)
Domain + Email (optional)	Domain registration / transfer, DNS management, basic email forwarding or MX.	Reseller API or full registrar stack + CIS DNS

6. Technical Implementation Steps (Detailed)

6.1 Prerequisites & Account Setup

- IBM Cloud account with sufficient quota (VPC, ROKS, CIS, Object Storage, IAM).
- Existing ROKS cluster (or provision a new dedicated one for hosting plane).
- IBM Cloud Internet Services (CIS) instance — create if not already present.
- Domain(s) you control (e.g. i3technologies.co.ke or a new hosting brand domain).
- oc / ibmcloud CLI, Git, ArgoCD access (as documented in SysAdmin guide).

6.2 Static Website Hosting with IBM Cloud Object Storage

Best for pure static sites. Steps (high level):

1. Create a COS bucket, enable public access / static website hosting, set index and error documents.
2. Upload site content (CLI, SDK, or CI pipeline).
3. Optionally front the bucket with CIS for custom domain, HTTPS, caching, WAF.
4. Point customer domain (CNAME or full zone) to the CIS / COS endpoint.

6.3 Dynamic / Container Hosting on OpenShift (Recommended Core Path)

For each new customer (or partner):

1. **Create namespace** (or project): `oc new-project customer-<id>` or declarative YAML applied via ArgoCD.
2. Apply ResourceQuota + LimitRange + NetworkPolicy (include the openshift-dns egress rule already used in the platform).
3. Deploy the workload (Deployment, Service, Route or Ingress). Prefer GitOps: store tenant manifests under a tenants/ tree and let ArgoCD ApplicationSet create Applications.
4. **Custom domain:**
 - Create an OpenShift Route with the customer hostname, or
 - Use ibmcloud ks ingress domain create / CIS to manage the domain and certificate lifecycle.
 - Customer creates a CNAME from their domain to the IBM-provided or CIS hostname (or you manage the full zone).
5. TLS: Use OpenShift-managed certificates, cert-manager, or CIS-managed certificates.
6. Secrets: Follow the existing patch pattern (never commit real secrets). Use sealed-secrets, External Secrets Operator, or OpenBao if you extend the current unseal model.

6.4 Example Route + Custom Domain Pattern

After the customer points example.com (or www) via CNAME to the cluster's router / CIS endpoint, create a Route similar to:

```
apiVersion: route.openshift.io/v1
kind: Route
metadata:
  name: customer-site
  namespace: customer-acme
spec:
  host: www.customer-domain.com
  to:
    kind: Service
    name: web
  tls:
    termination: edge
  insecureEdgeTerminationPolicy: Redirect
```

6.5 CIS Integration (Critical for Production Hosting)

IBM Cloud Internet Services (powered by Cloudflare edge) provides:

- Authoritative DNS and easy CNAME / A record management
- Global Load Balancer with health checks and geo-steering
- CDN / caching (page rules)
- Web Application Firewall (managed + custom rules)

- DDoS protection
- TLS / certificate management

Workflow: Add customer domain to CIS → update nameservers at the registrar (or keep CNAME-only) → create origin pool pointing at the OpenShift router / load-balancer hostname → attach Global Load Balancer → enable proxy mode for WAF/CDN.

6.6 Provisioning Automation (GitOps First)

Extend the existing ArgoCD + platform/ GitOps model:

- Tenant definition as YAML (or CRD) in a central Git repo.
- ArgoCD ApplicationSet that generates one Application per tenant.
- Optional: simple internal portal or API that writes the tenant YAML and opens a PR / commits (with approval).
- Builds: reuse Binary S2I or BuildConfig patterns already documented; or use Tekton / OpenShift Pipelines for customer Git repos.

7. Domain Services — Reseller vs Full ICANN Registrar

Offering domains is a natural upsell to web hosting. Two paths:

7.1 Recommended First Path — Domain Reseller

Partner with an existing ICANN-accredited registrar that offers a reseller / white-label API (many do). You sell domains under your brand, the upstream registrar handles registry interaction, WHOIS, escrow, and most compliance.

Advantages: Days to weeks to go live, low capital, minimal ICANN/ legal overhead, focus engineering effort on hosting platform.

What you still need: Reseller agreement, API integration (domain search, register, renew, transfer, DNS management), customer portal UI, billing integration, support process for domain disputes.

7.2 Full ICANN-Accredited Registrar Path

This is a formal accreditation process with ICANN. High-level steps and realistic expectations:

Step	Description	Notes / Cost
1. Eligibility & prep	Legal entity, financial stability (evidence of liquid capital, typically ~USD 70k+ working capital guidance), technical & operational plan, management team.	Legal + finance work
2. Application	Complete ICANN Registrar Accreditation Application + supporting documents. Non-refundable application fee.	~USD 3,500 application fee
3. Review	ICANN review (background checks, clarifications). Can take months.	3–9+ months typical
4. Agreements	Sign Registrar Accreditation Agreement (RAA). Sign Registrar Data Escrow (RDE) agreement with approved escrow agent.	Legal review required
5. Fees & listing	Pay annual accreditation fee. ICANN lists you and notifies registries.	Annual ~USD 4,000 (+ transaction fees)
6. Registry onboarding	Contract with each TLD registry (.com = Verisign, etc.). Security deposits, minimum commitments, technical certification (EPP, etc.).	Additional capital & tech work
7. Systems	Production-grade registration system (EPP client, WHOIS, RDAP, billing, abuse handling, data escrow deposits).	Build or buy (significant)

Recommendation for i3: Launch hosting first with a domain *reseller* partnership. Re-evaluate full accreditation only when domain volume and strategic control justify the cost and ongoing compliance burden.

8. Tools, Technologies & IBM Cloud Services Required

8.1 Core IBM Cloud Services

Service	Purpose in Hosting Offer
Red Hat OpenShift on IBM Cloud (ROKS)	Primary compute & orchestration platform (already running 4.17)
IBM Cloud Internet Services (CIS)	DNS, CDN, WAF, DDoS, Global LB, TLS — edge delivery for all customer sites
IBM Cloud Object Storage	Static website hosting, backups, media storage
IBM Cloud DNS Services (optional)	Private / internal DNS if needed
IBM Cloud Databases / existing PGO	Managed Postgres add-on for customers
IBM Cloud Container Registry	Private images for customer or platform workloads
IAM + Resource Groups	Access control, billing separation, least privilege
Activity Tracker / Log Analysis / Monitoring	Audit, centralised logs, metrics

8.2 Open-Source / Platform Tools (already in or easy to add)

- ArgoCD / ApplicationSets — GitOps tenant provisioning
- cert-manager or OpenShift cert management — TLS certificates
- External Secrets / Sealed Secrets / OpenBao — secret hygiene (extend current model)
- Prometheus + Grafana (+ existing Langfuse pattern) — observability
- Keycloak — identity (extend realm or multi-realm for customers/partners)
- Nginx / Caddy / Apache or language runtimes — application servers
- Velero or OADP — backup/restore of cluster resources and PVs

8.3 Domain-Related Tools

- Reseller API client (from chosen upstream registrar)
- Optional: full EPP client + registration platform if pursuing ICANN accreditation
- WHOIS / RDAP services and abuse contact handling

8.4 Developer / Ops Tooling

- oc (4.17+) and ibmcloud CLI — already required
- Git + existing GitHub repo model
- Terraform / Schematics — for CIS, COS, additional ROKS clusters
- CI (GitHub Actions, Tekton, or IBM Continuous Delivery)

9. Security, Compliance, Isolation & Governance

Reuse and harden patterns already present in the i3 SysAdmin guide:

- **NetworkPolicy** — default-deny + explicit allow (including the openshift-dns egress fix for OVN).
- **ResourceQuota / LimitRange** — prevent noisy-neighbour and runaway builds (see existing BuildConfig resource patch pattern).
- **Pod Security** — restricted or baseline PSS; avoid anyuid except where absolutely required (ERPNext example in current guide).
- **Secrets** — never commit real values; use the documented oc patch stringData pattern or move to a proper secrets operator.
- **TLS everywhere** — edge (CIS) + Route edge or re-encrypt.
- **WAF + DDoS** — enable CIS proxy mode for all public customer domains.
- **Image security** — internal registry, image signing / admission if required, vulnerability scanning.
- **Audit** — OpenShift audit logs + IBM Activity Tracker; retain according to policy.
- **Data residency** — keep eu-de (Frankfurt) for EU/African customers as needed; document clearly.

For partners/customers that process regulated data, offer dedicated namespaces with stricter controls or dedicated clusters, and document shared-responsibility boundaries in contracts.

10. Operations, Monitoring, Billing & Support Model

10.1 Day-2 Operations

- Extend the existing “Quick Health Check” loop to include hosting namespaces.
- Per-tenant resource usage dashboards (Prometheus + Grafana).
- Automated certificate renewal monitoring.
- Backup schedule for databases and critical PVs (Velero / OADP + COS).
- Clear runbooks for the common issues already documented (RWO multi-attach, NetworkPolicy DNS, BuildConfig quotas, etc.).

10.2 Metering & Billing

Decide early how you will charge: fixed packages, usage-based (CPU/memory/storage/egress), or hybrid. Technical enablers:

- OpenShift / Kubernetes metrics + custom exporters
- IBM Cloud billing APIs / Cost and Usage reports for underlying infrastructure
- Application-level metering if you expose AI or other premium features
- Invoice generation and payment integration (separate finance system)

10.3 Support Model

- Define SLAs per tier (response & resolution times).
- Single pane for tickets (existing tools or new).
- Clear escalation to the current admin team (Sebastian, Philip, Joseph) for platform-level incidents.
- Customer-facing status page (optional but valuable).

11. Disaster Recovery & Business Continuity

Extend the priority order already defined in the SysAdmin guide:

1. Postgres HA (i3-data) and any customer databases
2. Keycloak / authentication
3. Model Gateway (if AI features are sold)
4. Edge (CIS) configuration & DNS (export / IaC)
5. Tenant application namespaces (GitOps re-apply)

Re-bootstrap procedure remains: oc login → apply platform (and tenants) from Git → re-patch secrets → re-pull models if needed → verify Routes and CIS. Keep Keycloak realm exports and database dumps on a regular schedule (manual pg_dump pattern already documented).

12. Commercial Model, Pricing Guidance & Go-to-Market

Exact pricing is a business decision; technical document only provides structure:

- Cost of goods: ROKS worker nodes, storage, egress, CIS plan, any managed DB, support labour.
- Package pricing with clear resource envelopes (CPU, memory, storage, bandwidth, number of sites).
- Domain registration as a low-margin or break-even lead-in product that increases stickiness of hosting.
- Partner program: higher discounts or white-label for agencies that bring multiple end customers.
- Contractual shared-responsibility matrix (what i3 manages vs what the customer manages).

13. Implementation Roadmap (Phased)

Phase	Scope	Outcome
0 — Foundation (2–4 weeks)	CIS instance, base NetworkPolicy/Quota templates, ArgoCD ApplicationSet skeleton, COS static hosting PoC, one pilot customer namespace.	Repeatable tenant create path
1 — MVP Hosting (4–8 weeks)	Static + simple container hosting, custom domain via CNAME + Route/CIS, basic portal or CLI for internal ops, monitoring & backup basics.	First paying / partner sites live
2 — Productisation (ongoing)	Self-service or semi-automated provisioning, WordPress/CMS flavour, database add-on, usage metering, support runbooks, SLA definition.	Sellable catalogue
3 — Domains (parallel or later)	Reseller API integration + UI; later evaluate full ICANN path if volume justifies.	Domain upsell available
4 — Scale & Enterprise	Dedicated node pools / clusters, advanced WAF rules, multi-region if required, formal compliance packaging.	Enterprise-ready offer

14. Known Limitations & Recommendations

- Current cluster is CPU-only and already runs AI workloads — protect capacity with strict quotas and consider a dedicated hosting cluster or node pool.
- RWO PVC multi-attach issues remain a common operational risk; document and automate the force-delete + scale-rs pattern.
- Single Ollama instance and model gateway have finite capacity; rate-limit any AI features sold to customers.
- Full ICANN accreditation is capital- and compliance-intensive; start as reseller.
- Keep solution-01...08 and production AI namespaces strictly out of the multi-tenant hosting plane.
- Legal / contracts / privacy policy / AUP / DMCA or local equivalent processes must be in place before accepting external customers.

15. Appendix — Useful Commands & References

15.1 Cluster Login (from existing guide)

```
oc login https://c113-e.eu-de.containers.cloud.ibm.com:32581 --token=<token-from-IBM-Cloud-console>
```

15.2 Create a Tenant Namespace Skeleton

```
oc new-project customer-acme  
# then apply Quota, LimitRange, NetworkPolicy YAMLs from your tenants/ templates
```

15.3 Rolling Restart / Common Fixes

Reuse the RWO multi-attach, NetworkPolicy DNS, and BuildConfig resource limit fixes already documented in the i3 SysAdmin Guide sections 3 and 9.

15.4 Key External References

- IBM Cloud Object Storage — Static website hosting documentation
- IBM Cloud Internet Services (CIS) documentation — DNS, GLB, WAF, CDN
- Red Hat OpenShift on IBM Cloud — Routes, Ingress, custom domains
- ICANN — “How to become a registrar” and Registrar Accreditation Agreement
- Existing internal: i3 AI Platform System Administrator Guide (cafb76f)

Document prepared for i3 Technologies leadership and technical teams. Built on the production i3 AI Platform running on IBM ROKS 4.17 (eu-de). For questions contact the platform admin team: Sebastian Njagi · Philip Mukiti · Joseph Maina.

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